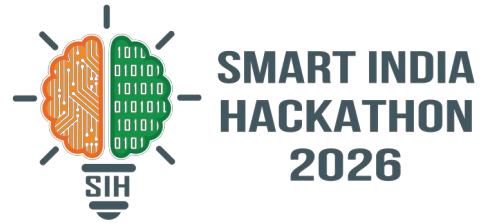


SMART INDIA HACKATHON 2026



TITLE PAGE

CryptoDrishti

An evidence-driven tool for discovering enterprise cryptography, assessing quantum exposure and planning post-quantum migration.

Problem Statement ID - SIH26164

Problem Statement Title - Enterprise Cryptographic Discovery & Analysis Tool (ECDAT)

Theme - Blockchain & Cybersecurity

PS Category - Software

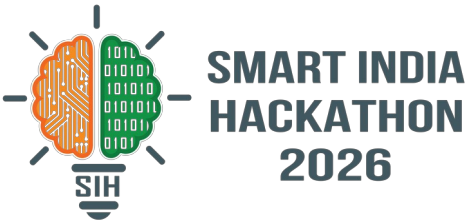
Team ID - 146876

Team Name - Zero-Day





CryptoDrishti | PROPOSED SOLUTION



The problem

Cryptographic assets are spread across application code, third-party libraries, configuration, certificates, compiled binaries and container images. Each records cryptography differently, and none of them is a list. Without a reliable inventory, an organisation cannot say which assets need quantum-safe migration, or in what order.

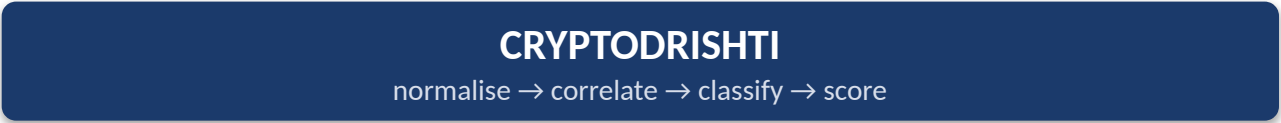
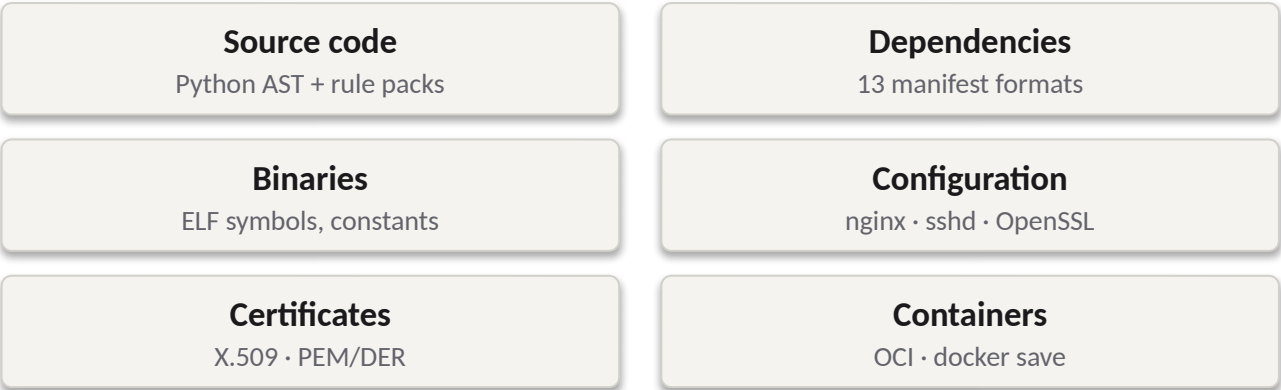
What CryptoDrishti does

CryptoDrishti discovers cryptographic evidence across those surfaces, normalises it into one inventory, assesses each asset's quantum exposure, and produces evidence-backed migration recommendations and a CycloneDX Cryptography Bill of Materials.

Why it is different

Finding RSA in a library is not observing RSA signing at a call site — capability, declaration, use and observation stay apart.

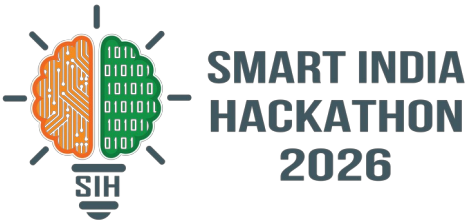
WHERE THE EVIDENCE COMES FROM



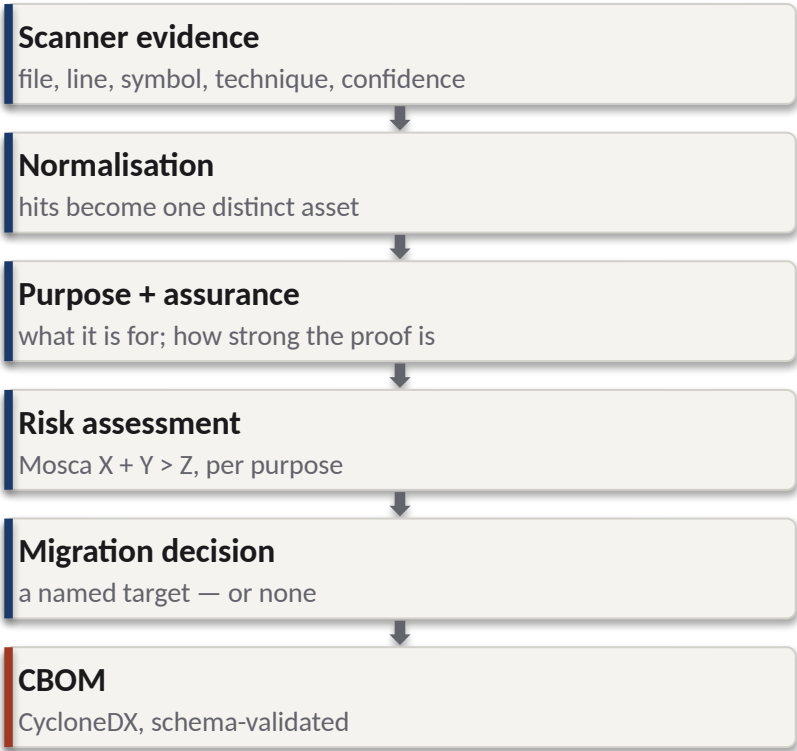
A seventh sensor completes a live TLS handshake — only against an endpoint the operator names, and only after the destination policy vets every resolved address.



TECHNICAL APPROACH



How a finding is processed



ONE ALGORITHM • THREE PURPOSE STATES • THREE DECISIONS

Every scanner can tell you RSA is present. What decides the migration is what it is being used for.

Case 1 — RSA signing
svc-payments/signing.py:15 · source · used

RSA-PSS padding signs

ML-DSA-65

Case 2 — RSA key establishment
svc-gateway/transport.py:16 · source · used

RSA-OAEP padding wraps a key

X25519MLKEM768

Case 3 — RSA, unresolved
legacy/nginx.conf:5 · config · declared

a cipher list permits RSA, purpose unstated

Purpose must be resolved first

BUILT WITH

Python 3.11+ · FastAPI

SQLite (WAL)

Vanilla JS console

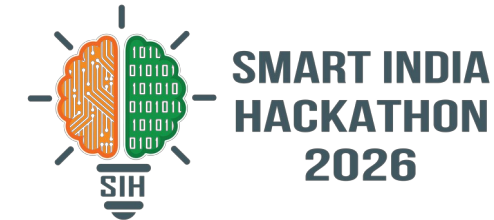
60 rules · 8 languages

ELF symbol parsing

CycloneDX 1.6 / 1.7



FEASIBILITY AND VIABILITY



Implemented

- Seven scanning surfaces, each returning evidence with a location, a technique and a confidence
- An inventory of distinct assets, carrying purpose and assurance on every one
- A quantum-risk model taking operator-supplied shelf life, criticality and constraints, saved per asset and surviving a rescan

Validated

- **664 automated tests** — on Python 3.11, 3.12 and 3.13 in CI
- **Official schema validation** — every CBOM checked offline against the CycloneDX schemas at pinned commit 0bd48c8; CI fails on a violation
- **Hand-labelled benchmark** — 116 findings over six scanners, labelled from the fixtures, not from the tool's output

Feasible to deploy

The tool is local-first: one machine, loopback by default, no cloud service, and it validates its own output offline. Committed demo fixtures reproduce every figure in this deck with one command.

Limits — Full AST analysis is Python only; seven further languages use rule packs, and Rust and Swift are recognised but effectively uncovered. Binary parsing is ELF only. No KMS, HSM, cloud or Kubernetes integration. A working prototype, not a deployed product.

PARTIAL SCAN — this inventory is incomplete.

1 endpoint(s) refused by network policy
endpoint refused: 169.254.169.254 — 169.254.169.254 resolves only to addresses this tool must not probe — 169.254.169.254: cloud instance metadata service. Set CD_ALLOW_PRIVATE_TARGETS=1 only for a network you are authorized to test.

Findings from the sensors that did run are valid. Absence of a finding is not evidence of absence.

9 assets break outright under Shor's algorithm, and **7** more lose half their strength to Grover. **6** are already quantum-safe.

ASSETS	23
distinct cryptographic assets	
VULNERABLE	16
broken or weakened by quantum attack	
CRITICAL	2
risk score at or above 70	
PEAK EXPOSURE	19.7
years readable past Q-Day	

Assessment — **PARTIAL** stated first, then 23 assets, 16 vulnerable.

rsa CLOSE

Asymmetric padding scheme

■ Broken by Shor critical · 100.0 USED production

WHAT THE EVIDENCE ESTABLISHES

Purpose signature

Assurance used

Proves use yes

Purpose RSA-PSS padding is a signature scheme.

Code invokes it — a resolved call site or a linked binary symbol.

Evidence drawer — one finding's proof.

MOSCA EXPOSURE

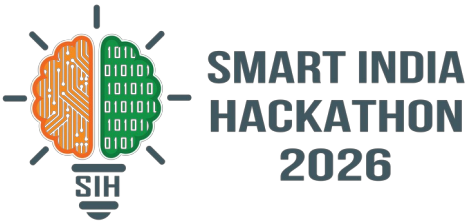
Secrecy (X)	25 yr
Migration (Y)	2 yr
To Q-Day (Z)	7.27 yr
Exposure	19.7 yr
P(exposed)	100%

Critically exposed. This key will still be trusted, and forgeable, well past the scenario's arrival window.

The exposure arithmetic behind its score.



IMPACT AND BENEFITS



What a migration team does with the output



Intended for security engineering teams, cryptographic inventory and migration-planning teams, and infrastructure operators. These are intended users; the tool has not been deployed in production.

What changes for them

- “We think we use RSA somewhere” becomes a per-asset list with file, line, purpose, evidence strength and a named replacement.
- What an estate runs is separated from what it merely has installed, so the plan is not padded with libraries nobody calls.
- A partial scan is labelled partial, so nobody signs off an estate on an inventory that silently missed half of it.

Measured on a labelled corpus — like-for-like, before → after

F1 0.941 → 0.983

79/5/5 → 84 TP · 3 FP · 0 FN

An expanded synthetic corpus of 116 findings across six scanners scores 1.000 — a separate, non-comparable experiment. Both corpora were written by this project's developers and measure those corpora only; real-world enterprise accuracy has not been measured.

What it generates

SIH26164			
Cryptographic Risk Assessment			
demo-estate · generated 24 September 2026, 21:29 by CryptoDrishti 0.9.0			
CRYPTOGRAPHIC ASSETS	QUANTUM VULNERABLE	CRITICAL	PEAK EXPOSURE
23	16	2	19.7
from 30 detector observations	69.6% of the estate	6 high severity	years readable after Q-Day

The generated report — headline counts from the same scan.

Structural check: PASS (23 components). Official CycloneDX 1.6 (ECMA-424) JSON Schema: PASS (pinned 0bd48c88).

CBOM validation, run live in the console — both checks pass.

Cryptography standards informing classification and migration

- FIPS 203 — Module-Lattice-Based Key-Encapsulation Mechanism (ML-KEM). NIST, 2024. doi.org/10.6028/NIST.FIPS.203
- FIPS 204 — Module-Lattice-Based Digital Signature Standard (ML-DSA). NIST, 2024. doi.org/10.6028/NIST.FIPS.204
- FIPS 205 — Stateless Hash-Based Digital Signature Standard (SLH-DSA). NIST, 2024. doi.org/10.6028/NIST.FIPS.205
- NIST IR 8547 (initial public draft) — Transition to Post-Quantum Cryptography Standards. NIST, 2024.

Problem statement and repository

- Smart India Hackathon 2026, Problem Statement SIH26164 — Enterprise Cryptographic Discovery & Analysis Tool (ECDAT), National Technical Research Organisation (NTRO).
- Source, 664 tests, the labelled benchmark corpus and its committed results, and the vendored schemas: github.com/sgtsujith141-wq/cryptodrishti

CBOM and container formats

- CycloneDX v1.6, standardised as ECMA-424 (1st edition, June 2024) — cyclonedx.org/specification/overview
- CycloneDX v1.7 JSON Schema. Both schemas vendored in this repository at pinned commit 0bd48c8 and checksummed.
- OCI Image Format Specification — image layout, manifests and layer whiteouts. github.com/opencontainers/image-spec
- RFC 8996 — Deprecating TLS 1.0 and TLS 1.1. rfc-editor.org/rfc/rfc8996

Risk methodology

- M. Mosca, “Cybersecurity in an Era with Quantum Computers: Will We Be Ready?”, IEEE Security & Privacy 16(5), 2018 — the source of the $X + Y > Z$ inequality.
- P. W. Shor, “Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms”, SIAM J. Computing, 1997.
- L. K. Grover, “A fast quantum mechanical algorithm for database search”, STOC 1996.

Repository

github.com/sgtsujith141-wq/cryptodrishti

Demo video

[YOUTUBE LINK TO BE ADDED AFTER UPLOAD]

CycloneDX 1.6 / 1.7 — validated

offline, against the official JSON Schema at a pinned commit



Scope — CryptoDrishti discovers, classifies, assesses and recommends cryptography. It does not implement the post-quantum primitives above; it uses them as the standards that inform classification and migration targets.